

Large-Scale Associations of BMI and Clinical Obesity With Cardiac Field Artifact in Scalp EEG

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Abstract

Heartbeat-evoked potentials (HEPs) are used to study cardiac interoception; however, the cardiac electrical field is time-locked to the same R-peak and contaminates scalp EEG. Previous studies have characterised this cardiac field artifact (CFA) in small, single-cohort samples and implicitly treated residual contamination after QRS exclusion as constant; yet the extent to which CFA varies with BMI, medical conditions, sex, and age, and whether short segments underestimate it, have not been examined. To address this gap, we quantified CFA in 7,995 polysomnography patients (98,203 channel-recordings) from the Human Sleep Project, a publicly available, curated dataset. Per channel, the R-peak-locked EEG average was regressed on the ECG average (-300 to 400 ms), excluding a ± 50 ms QRS interval. Outside the QRS interval, the ECG explained a mean R^2 of 0.32 (full epoch: 0.35), and removing the ECG-related independent component reduced HEP variance by a median of 28%. CFA varied across electrodes and was greater in male than in female patients (0.37 vs. 0.32) and in patients with than without a linked diagnosis (0.35 vs. 0.27; confounded by recording site); among diagnoses, it was highest for obesity. BMI correlated weakly with CFA (exploratory; $r = 0.11$); age showed no linear trend. In 7,079 duration-matched patients, mean R^2 increased from 0.30 at 5 min to 0.42 at 60 min, suggesting that short segments underestimate contamination. CFA is a participant- and channel-dependent confound. HEP studies should therefore model ECG-derived contamination per channel and account for BMI, clinical obesity, and sex when comparing groups.

Keywords: cardiac field artifact; heartbeat-evoked potential; interoception; independent component analysis; EEG-ECG volume conduction; population neuroscience; polysomnography

1. Introduction

The electrical field generated by each heartbeat is volume-conducted to scalp EEG electrodes with a fixed phase relationship to the R-peak, the same event used to time-lock heartbeat-evoked potential (HEP) epochs. This cardiac field artifact (CFA) is therefore preserved by trial averaging to the same degree as genuine cortical interoceptive activity.¹ The standard mitigation is to exclude a short interval around the QRS complex (typically ± 30 –50 ms) from analysis.^{2–4} Previous studies have characterised the topography and morphology of CFA in single cohorts of a few tens of participants,^{1,5} samples underpowered to detect modest between-participant differences; yet, to our knowledge, no study has quantified the proportion of HEP variance that remains shared with the ECG after QRS exclusion at the population level, nor examined whether this proportion varies with BMI, medical conditions, sex, and age, or with the length of the analysed segment. Instead, residual contamination after QRS exclusion has largely been treated as an approximately constant property of the recording.

The assumption of constant contamination matters because the R-peak-locked scalp waveform is a mixture of sources rather than a measurement of a single generator. Neural responses to baroreceptor and somatosensory input coexist with passive cardiac volume conduction, and both are preserved by heartbeat-locked averaging.^{2,5,6} A group difference in HEP amplitude may therefore reflect cortical processing, cardiac electrophysiology, tissue conductivity, electrode geometry, or a combination of these factors. Recent reviews have documented considerable heterogeneity in how CFA is handled and have noted that inconsistent preprocessing limits reproducibility and clinical interpretation.^{3,4,7} Quantifying the residual ECG-related variance is therefore a prerequisite for interpreting HEPs as neural biomarkers, not merely a technical refinement.

CFA may also differ systematically between individuals. Body composition alters the geometry and conductivity of the path between the heart and the scalp. In surface ECG, subcutaneous fat attenuates cardiac voltage, and correcting voltage for measured body fat alters its relationship with cardiac structure and ambulatory blood pressure.⁸ Clinical obesity and BMI are therefore of particular interest, although neither directly measures body-fat distribution. Sex, age, and disease burden also covary with body composition, cardiac morphology, rhythm, and medication exposure. If body composition or these characteristics predict CFA, they constitute potential confounders in between-group HEP analyses and should be measured or modelled rather than assumed to be eliminated by a fixed QRS exclusion.

Here, we addressed these questions in clinical polysomnography recordings from the Human Sleep Project, a publicly available, curated dataset,⁹ using two complementary estimators applied to a common HEP epoch window. A model-free estimator regresses each channel's R-peak-locked evoked average on the same patient's R-peak-locked ECG evoked average and therefore requires no assumption that ICA correctly separates cardiac from neural sources. A model-based estimator applies ECG-informed ICA and quantifies both the share of HEP variance carried by the ECG-related component and the variance reduction obtained by removing it. To our knowledge, this 7,995-patient cohort is the largest in which the contribution of CFA to HEP variance has been quantified. We additionally examined whether segment duration affects the apparent contamination, as short segments may contain too few heartbeats to yield a stable evoked average. We hypothesised that ECG-related variance would persist outside the QRS interval, would vary across channels and patient characteristics, and would increase with segment length as the shared signal became more stable.

2. Methods

2.1 Cohort and recordings

This study is a secondary analysis of existing polysomnography recordings; no recordings were acquired for this study. Patient data were taken from the Human Sleep Project,⁹ a publicly available, curated dataset of clinical polysomnography recordings with linked electronic health records (EHR), distributed through the Brain Data Science Platform. BMI was computed from each patient's median height and weight recorded in the EHR vitals and admission tables; values outside 10–80 kg/m² were excluded as implausible. EEG channels were identified by matching channel labels against standard 10-20/10-10 electrode names, thereby excluding intracranial depth electrodes and auxiliary/DC channels; at least two

EEG channels were required. The ECG channel was identified by label pattern matching (ECG/EKG), and the first matching lead was used. Cohort characteristics are reported in §3.1.

2.2 Window selection, R-peak detection, and preprocessing

For each recording, a single 10-min analysis window was selected by a seeded, reproducible random search (uniformly distributed start times; up to 10 attempts). R-peaks were detected on the ECG of each candidate window before the analysis filter described below: the median-subtracted signal was band-pass filtered at 5–25 Hz (third-order zero-phase Butterworth), rectified, and peaks were identified with a minimum inter-peak distance of 300 ms and a prominence threshold of three times the median absolute deviation of the rectified signal; rectification made detection insensitive to ECG polarity. A window was accepted if (i) at least 70% of EEG channels passed signal-quality criteria ($\geq 99.9\%$ finite samples; standard deviation 0.1–500 μV ; peak-to-peak amplitude $\leq 5,000 \mu\text{V}$; $< 20\%$ flat samples), (ii) the mean heart rate was 35–180 beats/min, and (iii) at least 70% of R-R intervals lay between 0.33 and 2.0 s. The accepted window was then band-pass filtered between 1 Hz and $\min(100 \text{ Hz}, \text{Nyquist} - 0.5 \text{ Hz})$ using a zero-phase FIR filter in MNE-Python;¹⁰ because 86% of recordings were sampled at 200 Hz, the upper edge lay just below the Nyquist frequency in most recordings. The same filter was applied to the EEG and ECG channels, and signals were analysed in μV at their native sampling rate without re-referencing, channel interpolation, or amplitude-based rejection, so that CFA was estimated in minimally processed data.

2.3 Model-free CFA estimator: HEP-vs-ECG regression

HEP epochs spanned -300 to 400 ms relative to each detected R-peak; epochs extending beyond the window edges were discarded, no baseline correction was applied, and recordings with fewer than 20 epochs were excluded (median 940 epochs per 10-min window). For each patient and EEG channel, the R-peak-locked evoked average (mean across epochs) was computed, together with the evoked average of the patient's ECG lead over the same epoch window. The squared zero-lag Pearson correlation (R^2) between the two evoked waveforms was taken as the fraction of HEP variance in that channel shared with the ECG and used as an index of CFA. R^2 was computed over the full epoch and over the samples outside a ± 50 ms QRS-exclusion window, as is standard in HEP analysis.⁴ This estimator does not depend on the ability of ICA to isolate a cardiac source; rather, it tests directly the extent to which the averaged scalp waveform is a linearly scaled (and offset) copy of the averaged ECG. For patient-level analyses (§2.7), channel-level R^2 values were averaged across each patient's channels at the six well-covered sites (§2.5) and, for patients with more than one recording, across recordings (patient-mean CFA R^2).

2.4 Model-based CFA estimator: ECG-informed ICA

In parallel, ICA¹¹ (extended Picard algorithm¹²; $\min(15, \text{number of EEG channels} - 1)$ components; 500 iterations; fixed random seed) was fitted to the filtered EEG of the same 10-min window used for the regression estimator. Components whose correlation with the ECG channel exceeded the default threshold were identified as ECG-related using the correlation-based scoring implemented in MNE-Python;¹⁰ if none did, the component with the highest absolute score was used (two components were flagged in 49 of 13,624 recordings). Rather than each component's share of continuous-signal variance, its mixing-weighted contribution to each channel was evaluated on the R-peak-locked evoked average, thereby quantifying the fraction of heartbeat-evoked signal that the component explains. In addition, the effect of artifact removal was measured directly by comparing the HEP variance of each channel before and after excluding the ECG-related component through ICA back-projection, yielding the realised percentage reduction in variance, $100 \times (\text{pre} - \text{post})/\text{pre}$, rather than the nominal share of the component. For Figure 1b, the ratio of ECG-related component variance to residual variance (SNR) was also computed for each channel-recording.

2.5 Channel canonicalisation and minimum coverage

Channel labels were mapped to their scalp-side 10-20 site, and labels corresponding only to a reference electrode were excluded. Figure 2 includes all canonical sites recorded in at least 500 patients. Analyses requiring comparison across conditions or estimators at a common set of electrodes (§3.3–3.4, S2–S4) were restricted to sites present in at least 50%

of the patients included in that analysis. In practice, this criterion retained six sites (F3, F4, C3, C4, O1, O2; each present in 96–100% of patients), which constitute the predominant montage of the dataset.

2.6 EEG-ECG cross-correlation and mutual information

The EEG-ECG relationship was further characterised in a separate subsample of 150 patients (Supplementary Figures S3–S4; 148 with usable cross-correlation and mutual-information data). Lag-resolved cross-correlation was computed as the Pearson correlation between each channel's evoked waveform and the concurrent ECG evoked average at lags of ± 100 ms in 5 ms steps, extending the zero-lag analysis of §2.3; the peak absolute correlation and its lag were reported after per-channel sign alignment, as reference polarity is arbitrary. Mutual information was estimated with the Kraskov k-nearest-neighbour estimator¹³ to capture both linear and nonlinear EEG-ECG dependence. Both measures were computed for three conditions: pre-ICA, post-ICA, and a non-heartbeat-locked control in which the same window was re-epoching around an equal number of pseudo-events placed uniformly at random instead of R-peaks; in this subsample, ICA was refitted and R-peaks were re-detected on the band-pass-filtered ECG. For the power spectral density (PSD) comparison (Figure S4), the patient's ECG evoked average was added as a fourth condition, and Welch PSD (dB) was computed for each condition at the six well-covered electrodes (§2.5). Two further random subsamples supported Figure S2: in one (973 patients), the ICA windows were re-epoching around the same number of pseudo-events, without ICA refitting, to estimate a variance noise floor; in the other (492 patients), ICA was refitted and the spectral entropy of each evoked waveform (Shannon entropy of its Welch power spectrum, normalised to 0–1) was computed for the pre-ICA, post-ICA, and pseudo-event conditions.

2.7 Statistics

Diagnoses were assigned to 15 predefined categories by case-insensitive keyword matching on EHR diagnosis descriptions (Supplementary Table S1); a patient could belong to several categories. Patients were classified as having a linked diagnosis if any diagnosis was recorded in the EHR, and as having no linked diagnosis if they were EHR-linked but had no recorded diagnosis (a group that, owing to incomplete diagnosis linkage, included patients from a site without diagnosis tables; see Limitations); patients without an EHR link were excluded from diagnosis analyses. The association between age and patient-mean CFA R^2 (outside QRS) was assessed as a continuous variable using Pearson correlation and by tertile using one-way ANOVA. Because the R^2 distribution was right-skewed, differences by sex and by presence of a linked diagnosis were assessed with the Mann-Whitney U test. Differences among diagnosis categories (§3.3) were assessed with the Kruskal-Wallis test followed by pairwise Mann-Whitney tests, with Benjamini-Hochberg false discovery rate (FDR) correction¹⁴ across all pairwise comparisons. The association with continuous BMI, an exploratory (not prespecified) analysis, was assessed using Pearson correlation. To retain one observation per patient, the BMI analysis used the longest available CFA window for each patient (30 min for 98% of patients), so absolute R^2 values in the BMI and BMI-adjusted sex analyses are not directly comparable with the 10-min estimates. BMI correlations were also estimated separately by sex, and a BMI-by-sex interaction term in an ordinary least-squares (OLS) model was used to test whether the slopes differed. A supplementary OLS model compared the sex coefficient before and after adjustment for BMI in the same subset. The relationship between model-free R^2 and the log-transformed ICA SNR (§2.4) was assessed with Pearson correlation across channel-recordings (Figure 1b). Sensitivity to segment duration (Supplementary §S6) was assessed in patients with usable data at all six durations (5–60 min), using all EEG channels rather than only the six well-covered sites (Figure S6a reports channel-level means), comparing sexes and diagnosis groups with Mann-Whitney tests at each duration. Windows of each length were drawn with the same seeded procedure, so that shorter windows were usually nested within longer ones; the 45- and 60-min analyses used one recording per patient. Sample sizes differ between analyses because each requires different data: the model-free estimator included all patients with a valid window (7,995); the ICA estimator, patients with a converged decomposition (7,933); age, sex, and diagnosis analyses, EHR-linked patients with known age and at least one well-covered site (7,729); BMI analyses, patients with a plausible BMI (6,127); and the duration analysis, patients with data at all six durations (7,079).

3. Results

3.1 Cohort

Demographic data were available for 7,732 of the 7,995 analysed patients (median age 61 years, range 7–103; Male $n=3,965$, Female $n=3,767$). This EHR-linked cohort consisted of clinically referred polysomnography patients with a high diagnostic burden rather than a healthy community sample; the most prevalent diagnosis categories were Obstructive Sleep Apnea ($n=5,524$); Hypertension ($n=3,882$); Obesity ($n=3,016$) (categories are not mutually exclusive). Cohort composition is shown in Supplementary Figure S1.

3.2 Model-free and ICA-based CFA estimates

Figure 1. Model-free and ICA-based cardiac-field-artifact (CFA) variance explained ($n = 7,995$ patients, 98,203 channel-recordings)

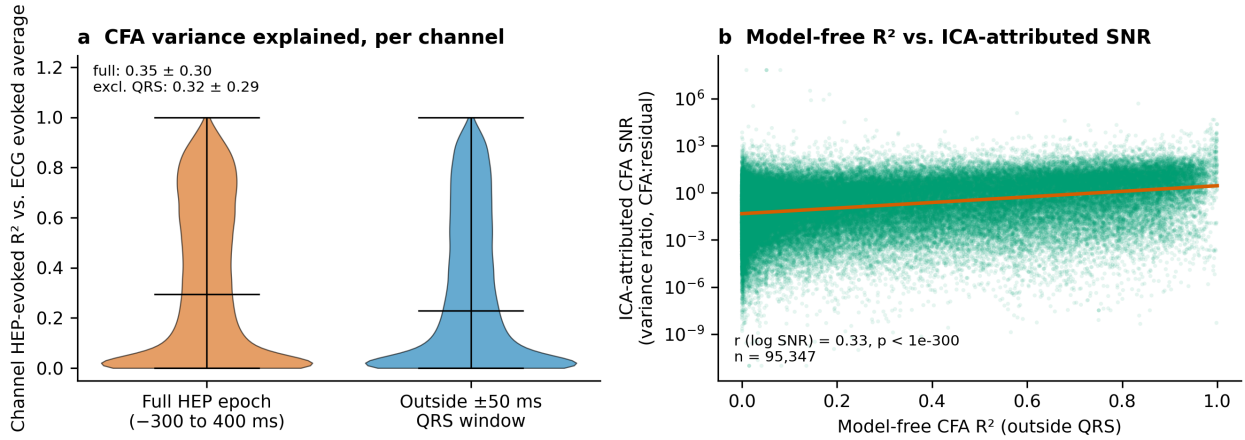


Figure 1. Cardiac field artifact (CFA) estimates from two complementary estimators across 7,995 patients (98,203 channel-recordings). (a) Model-free: distribution of per-channel R^2 between the HEP evoked average and the ECG evoked average, over the full epoch (mean 0.35) and outside the ± 50 ms QRS-exclusion window (mean 0.32). (b) Model-based estimator (7,933 patients): ICA SNR (ECG-related component variance relative to residual variance; §2.4) plotted against the model-free CFA R^2 of the same channel-recording (all channels; logarithmic SNR axis; line, least-squares fit of log SNR on R^2 ; $r = 0.33$, $p < 1e-300$, $n = 95,347$). No channel-level variance threshold was applied.

Outside the conventional QRS-exclusion window, the ECG evoked average explained a substantial proportion of HEP variance (mean $R^2 = 0.32$, median 0.23). This value was modestly lower than the full-epoch estimate (mean $R^2 = 0.35$), indicating that QRS exclusion removed only approximately 10% of the ECG-explained variance. In the 148-patient subsample with a chance-level reference (§2.6), full-epoch R^2 at F3/F4/C3/C4 was 0.35 for R-peak-locked averages but 0.04 (median 0.03) for averages around random pseudo-events, so the observed R^2 far exceeded chance similarity between finite-length evoked waveforms.

Across all channels (no loading threshold applied), the ECG-related component accounted for a median of 21% of HEP variance. The realised effect of removal was larger: excluding this component reduced HEP variance by a median of 28%. The dispersion in Figure 1b is expected, because a single, globally selected ICA source does not load equally on all electrodes, and because source variance and the change after back-projection are related but distinct quantities. We therefore regard the per-channel ECG regression as the primary estimate and the ICA-based removal as convergent evidence.

Figure 2. Where on the scalp is cardiac-field variance concentrated? (n = 7,916 patients, 89,951 channel-recordings, 15 sites)

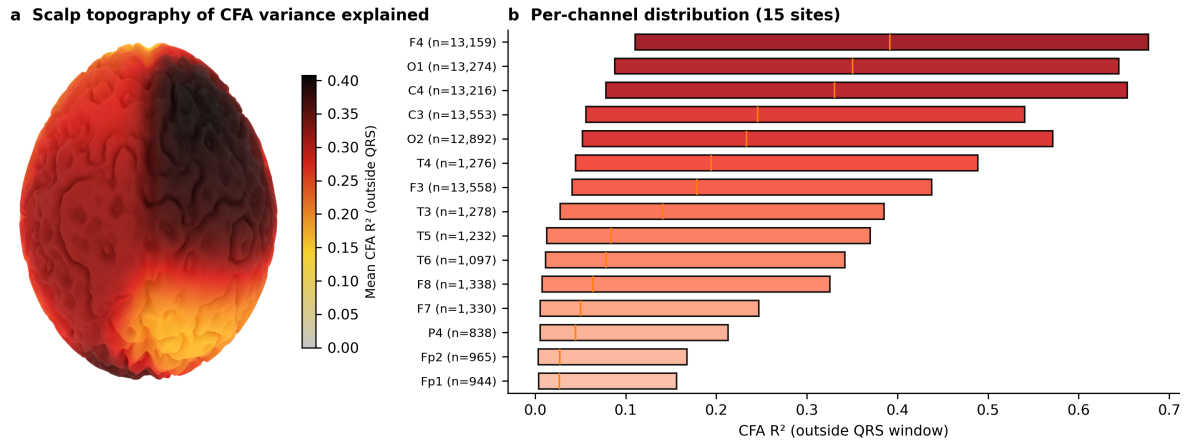


Figure 2. Scalp distribution of CFA R² (15 canonical 10-20 sites with at least 500 patients, 89,951 channel-recordings; bipolar/mastoid-referenced channel labels were canonicalised to their scalp-side site; Cz, Fz, Oz, P3, Pz were dropped for falling below the 500-patient floor). (a) Topographic map of mean CFA R² (outside QRS) per site. (b) Distribution of per-channel R² per site, sorted by median (orange line); box, interquartile range; n per site is indicated. Coverage is uneven, ranging from the six standard montage sites (F3/F4/C3/C4/O1/O2; n > 12,000 each) to sites near the 500-patient threshold, for which estimates are less precise.

CFA was not uniformly distributed across the scalp; it was highest at F4 (mean R² = 0.41) and lowest at Fp1 (mean R² = 0.12), an approximately 3.3-fold range. Because sites outside the standard montage came from a minority of recordings with different montages and references, part of this range reflects montage and referencing rather than electrode position; within the six standard, mastoid-referenced sites, mean R² ranged from 0.27 (F3) to 0.41 (F4). A site-independent CFA correction would therefore be expected to perform unevenly across electrodes, which argues for channel-level assessment.

3.3 CFA R² by diagnosis category

Figure 3. CFA variance explained by diagnosis category
(Kruskal-Wallis across all groups $p = 1.2e-21$)

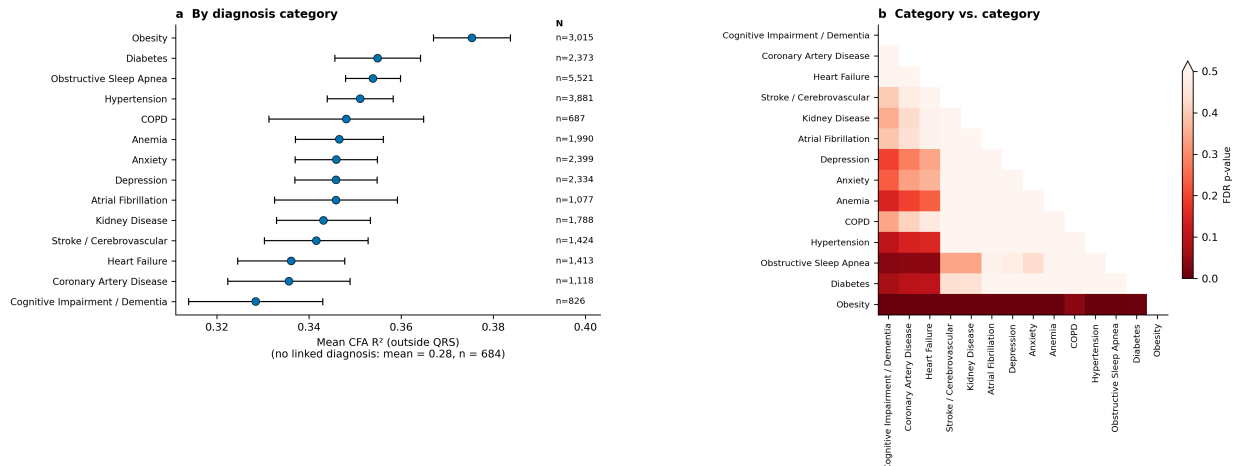


Figure 3. (a) Forest plot of patient-mean cardiac field artifact (CFA) R^2 (outside the QRS-exclusion window) per clinical diagnosis category; point = mean, error bar = 95% CI (Welch, unequal-variance). EHR-linked patients with no recorded diagnosis (reference; confounded by recording site, see Limitations) had mean CFA $R^2 = 0.27$ ($n = 500$); patients may belong to more than one diagnosis category. Categories are drawn from fifteen predefined diagnosis groups and sorted by mean; categories with fewer than 10 patients are omitted. Heart Transplant ($n = 133$; mean 0.31, 95% CI 0.27–0.35; Mann-Whitney $p = 0.053$ vs. the reference) is also omitted to improve axis resolution. Kruskal-Wallis test across all groups: $p = 2.4e-20$. (b) Pairwise comparisons among categories: Mann-Whitney p-values, Benjamini-Hochberg corrected across all 91 tests (colour scale: white, $q \geq 0.5$; red, $q = 0$; 16 pairs significant at $q < 0.05$); categories ordered as in (a).

All 14 displayed categories lay above the no-diagnosis reference, and all of these differences remained significant (Mann-Whitney) after FDR correction. Because these comparisons were unadjusted for age, sex, and BMI, and the reference group is confounded by recording site (see Limitations), they should not be interpreted as disease effects; the category-versus-category comparisons below are not affected by the site confound. Relative to the reference, the largest difference was observed in patients with Obesity (+0.11 R^2 units, $n = 3,015$) and the smallest in Cognitive Impairment / Dementia (+0.06, $n = 826$). Categories also differed from one another (Figure 3b): of 91 pairwise comparisons, 16 remained significant after FDR correction, of which 13 involved Obesity, which differed significantly from every other category; the remainder involved Obstructive Sleep Apnea.

3.4 CFA R^2 varies modestly with sex, age, and BMI

Figure 4. CFA variance explained shifts modestly with sex, age, and BMI

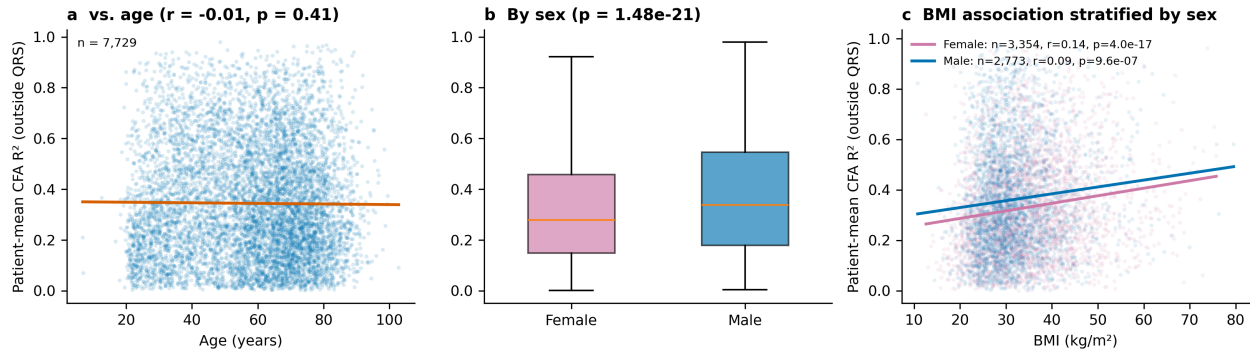


Figure 4. Patient-mean CFA R^2 (outside QRS) vs. (a) age, continuous ($n = 7,729$; Pearson $r = -0.01$, $p = 0.41$; line = least-squares fit), (b) sex (Mann-Whitney $p = 1.5e-21$), and (c) BMI, continuous ($n = 6,127$; Pearson $r = 0.11$, $p = 4.3e-17$; one observation per patient, longest available window; 5 min: $n=48$, 10 min: $n=22$, 20 min: $n=72$, 30 min: $n=5,985$; line = least-squares fit). Panel b: box shows quartiles; outliers omitted for clarity. Panel c shows sex-specific fits: female $n = 3,354$, $r = 0.14$, $p = 4.0e-17$; male $n = 2,773$, $r = 0.09$, $p = 9.6e-07$; BMI-by-sex interaction $p = 0.67$.

Patient-mean CFA R^2 was higher in male than female patients (0.37 vs. 0.32 , $p = 1.5e-21$), higher with a linked clinical diagnosis (0.35 vs. 0.27 , $p = 1.6e-15$; confounded by recording site, see Limitations), and slightly lower in the oldest age tertile than the younger two (0.33 vs. $0.35/0.35$, $p = 1.0e-04$). These effects were statistically significant but modest in magnitude (absolute differences of 0.02 – 0.05 R^2 units for sex and age) and were detectable owing to the large sample size. Age as a continuous variable showed no appreciable linear relationship with CFA R^2 (Figure 4a; Pearson $r = -0.01$, $p = 0.41$); tertile means were similar in the two younger tertiles and lower only in the oldest, a pattern that does not indicate a graded trend. In an exploratory analysis, BMI was positively associated with CFA R^2 (Figure 4c; $n = 6,127$, Pearson $r = 0.11$, $p = 4.3e-17$; mean BMI 33.5 kg/m^2). This association, although weak ($r^2 \approx 0.01$), is consistent with the obesity-diagnosis comparison (§3.3). The association was positive in both female patients ($r = 0.14$, $p = 4.0e-17$) and male patients ($r = 0.09$, $p = 9.6e-07$). The BMI-by-sex interaction was not significant ($p = 0.67$), providing no evidence that the BMI slope differs by sex.

4. Discussion

Consistent with our first hypothesis, two complementary estimators, a direct regression on the patient's ECG and an ICA decomposition with component removal, converged on the same conclusion: a substantial proportion of scalp HEP variance is shared with the ECG even after standard QRS exclusion. In line with the second hypothesis, this proportion was not a fixed property of the recording. It varied approximately 3.3-fold across scalp sites (partly reflecting montage and reference differences; §3.2), extending earlier topographic descriptions in small samples^{1,5} to the population level; it was higher in male than in female patients at every segment duration; and it was modestly lower in the oldest age tertile. The main-analysis difference between patients with and without a linked clinical diagnosis reversed within the site that had diagnosis records and was not reproduced in the duration-matched subset (Supplementary §S6; Limitations), so we do not interpret it further. Because the magnitude of CFA covaries with the variables around which many HEP group comparisons are organised, studies reporting sex-, BMI-, or diagnosis-related differences in HEP amplitude without channel-level CFA correction cannot exclude the possibility that part of the difference reflects cardiac-field contamination rather than cortical activity. The effect sizes are modest in absolute R^2 terms and do not by themselves invalidate larger HEP effects reported previously; they do, however, support routine channel-level CFA correction.

Although the artifact is most prominent around the QRS complex, the conventional ± 30 – 50 ms exclusion^{2–4} did not remove it: excluding ± 50 ms reduced mean ECG-shared variance (R^2) only from 0.35 to 0.32 , indicating that a substantial ECG-correlated component extends into the nominal analysis interval. The positive relationship between the model-free R^2 and the ICA-derived SNR suggests that both methods detect a common contamination process; channels

whose waveforms more closely resemble the ECG also contain more variance assigned to the ECG-related ICA source. Neither estimator establishes that every residual heartbeat-locked deflection is artifactual, and genuine cortical heartbeat-evoked activity is well documented.^{2,6} Nevertheless, the convergence of the two estimators indicates that uncorrected HEP variance cannot be assumed to be of cortical origin.

Clinically obese patients showed the highest mean CFA R^2 of all diagnosis categories and differed significantly from every other category (§3.3), consistent with a contribution of BMI or obesity-related characteristics to the conductive geometry that shapes the cardiac field. The exploratory continuous-BMI analysis points in the same direction (6,127 patients; $r = 0.11$). Because R^2 indexes waveform similarity rather than amplitude, these findings do not conflict with the attenuation of surface ECG voltage by subcutaneous fat;⁸ the underlying mechanism cannot be determined from the present data. The BMI–CFA association was numerically stronger in women than in men ($r = 0.14$ vs. 0.09); however, the BMI-by-sex interaction was not significant ($p = 0.67$), and this difference should not be interpreted as a sex-specific effect without replication. The sex difference itself persisted after adjustment for BMI (Supplementary §S5) and is therefore unlikely to be explained by BMI.

However, BMI is an anthropometric index rather than a direct measure of adiposity; no direct measures of fat mass, thoracic geometry, or electrode impedance were available, and diagnosis categories overlap. We therefore interpret these findings as evidence that BMI and clinical obesity are plausible confounders, not as evidence of a causal effect of body fat. Likewise, the associations of sex, age, and diagnosis category with CFA may reflect anatomy, cardiac physiology, medication, comorbidity, or recording conditions, and none can be assigned a specific biological interpretation on the basis of the present observational analysis.

Segment duration also affected the detectable contamination. In the duration-matched 7,079-patient analysis, mean R^2 rose from 0.30 at 5 min and 0.33 at 10 min to 0.42 at 60 min; the increase was already present between 5 and 30 min, where channel composition was nearly constant (§S6). Longer segments contain more heartbeats and yield a more stable evoked waveform, which presumably allows shared EEG-ECG structure to emerge from background noise; this pattern is consistent with our third hypothesis. Windows of 5 and 10 min, including the 10-min windows of the main analysis, therefore probably underestimate the detectable CFA burden in these data (see Limitations). This duration dependence does not identify a universally optimal segment length, but it indicates that HEP studies should justify the chosen segment length and assess the stability of their results across durations. In addition, because R^2 depends on the number of averaged heartbeats, between-group differences in heart rate could contribute to between-group differences in R^2 ; this was not examined in the present analysis.

In practice, EEG preprocessing for HEP analysis should combine concurrent ECG recording, channel-level assessment of the R-peak-locked EEG-ECG relationship, and removal or regression of ECG-related components, followed by verification that heartbeat-locked variance in the cleaned signal is reduced relative to the uncleaned data while remaining above a non-heartbeat-locked noise floor. Reporting only the QRS mask or the selected ICA component is insufficient. We recommend that studies report the analysis interval, the number of heartbeats, channel-level metrics before and after cleaning, and whether results are robust to adjustment for demographic and clinical variables that predict CFA, including BMI or, preferably, direct body-composition measures. These recommendations extend recent methodological reviews of HEP analysis and reporting.^{4,7}

5. Conclusions

Cardiac contamination is a substantial and systematic component of heartbeat-locked scalp EEG: the ECG explained a mean R^2 of 0.32 outside the QRS-exclusion window, and ICA-based removal of the ECG-related component reduced HEP variance by a median of 28%. The contamination varied with electrode, sex, obesity, and BMI, but only weakly with age; its apparent association with diagnostic burden was confounded by recording site. Estimated CFA also increased with segment duration up to the longest window examined (60 min). CFA should therefore be treated as a participant- and channel-dependent confound in HEP research. Robust inference requires ECG-informed channel-level cleaning with quantitative before-and-after validation, justified segment lengths checked for stability across durations,

and adjustment for BMI (ideally direct body-composition measures) and clinical characteristics. Without these safeguards, apparent group differences in neural responses may partly reflect the electrical field of the heart rather than cortical interoceptive processing.

Limitations

Several limitations should be noted. First, diagnosis records were available for only part of the Human Sleep Project, so the no-diagnosis reference group consisted largely of patients from a site without linked diagnosis tables (mean R^2 0.19, $n = 319$). Within the site with diagnosis records, patients without a recorded diagnosis had higher R^2 than those with one (0.41, $n = 181$, vs. 0.35, $n = 7,229$). The comparison of patients with and without a linked diagnosis is therefore confounded by recording site and should not be interpreted as a disease effect; comparisons among diagnosis categories are not affected by this confound. Second, the detected heart rate was high for sleep recordings (median 94 beats/min, interquartile range 78–111); the custom R-peak detector was not validated against an established algorithm, and occasional detection of T waves would mix T-wave-locked epochs into the averages. Third, R^2 does not distinguish volume-conducted CFA from neural activity whose time course resembles the ECG, and epochs ended at 400 ms after the R-peak, so the estimates do not cover later HEP intervals that overlap the T wave. Fourth, CFA was estimated in the native recording references, predominantly contralateral-mastoid derivations (e.g., F3-M2, F4-M1); because the cardiac field projects differently onto each reference electrode, site-level values and hemispheric asymmetries may not generalise to average-referenced HEP data.

Fifth, 10-min estimates are likely lower bounds with respect to segment duration, because mean CFA R^2 increased monotonically with duration, including from 5 to 30 min, where channel composition was nearly constant (§4; Supplementary §S6). Sixth, the chance-level (pseudo-event) R^2 was estimated only in a subsample of 148 patients and only over the full epoch (§3.2); a null for the outside-QRS estimate was not computed. Seventh, windows were not selected by sleep stage, so vigilance state was not controlled and may differ between groups and segment durations. Eighth, the cohort was predominantly a clinically referred population with a high diagnostic burden (Figure S1c), which limits the generalisability of absolute R^2 values to healthy volunteers; underweight patients were also under-represented (55 of 6,127 patients with measured BMI had a BMI <18.5 kg/m²). Finally, because this was a secondary analysis of existing datasets, recording parameters and clinical annotations could not be controlled by the authors.

Ethics statement

Human Sleep Project data were de-identified under the HIPAA Safe Harbor standard and made available under an approved institutional review board protocol (IRB protocol #2022P000417) with a waiver of informed consent; they were accessed under the Brain Data Science Platform data use agreement. No attempt was made to re-identify participants.

Data availability statement

The Human Sleep Project dataset is publicly available, subject to a data use agreement, from the Brain Data Science Platform (<https://bdsp.io/content/hsp/>). Analysis code is available from the corresponding author upon reasonable request (nircafri@mail.tau.ac.il).

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Declaration of competing interests

The authors declare no competing interests.

Supplementary analysis

S1. Cohort composition

Figure S1. Cohort composition (Harvard EHR-linked subset, n = 7,738)

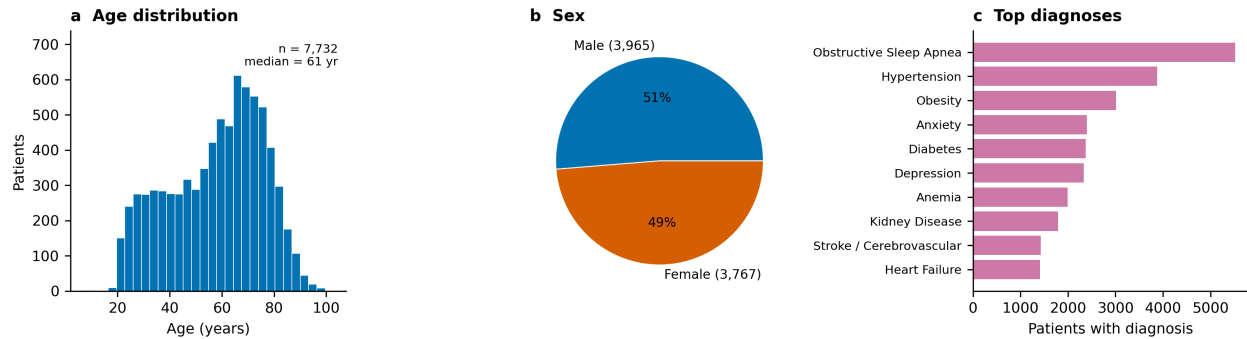


Figure S1. Cohort composition of the 7,738 Human Sleep Project patients. (a) Age distribution (median 61 years, range 7–103). (b) Sex (Male n=3,965, Female n=3,767). (c) The ten most prevalent clinical diagnosis categories, the largest of which were Obstructive Sleep Apnea (n=5,524); Hypertension (n=3,882); Obesity (n=3,016). The cohort is a clinically referred polysomnography population rather than a healthy community sample; diagnosis categories are not mutually exclusive.

Table S1. Diagnosis categories and the case-insensitive keywords matched against EHR diagnosis descriptions. Keyword matching is inclusive; for example, "infarction" also matches myocardial infarction.

Category	Keywords (any match)
Obstructive Sleep Apnea	sleep apnea
Hypertension	hypertension
Diabetes	diabetes
Heart Failure	heart failure
Coronary Artery Disease	coronary; atherosclerotic heart
Heart Transplant	heart transplant; transplant status
Atrial Fibrillation	atrial fibrillation; atrial flutter
COPD	chronic obstructive
Obesity	obesity; obese
Depression	depressive; depression
Anxiety	anxiety
Cognitive Impairment / Dementia	cognitive; alzheimer; dementia
Stroke / Cerebrovascular	infarction; cerebrovascular; stroke; hemorrhage
Kidney Disease	kidney; renal
Anemia	anemia

S2. Variance and entropy vs. a non-heartbeat-locked noise floor

Figure S3. HEP-evoked EEG variance and spectral entropy vs. a non-heartbeat-locked noise floor
(n = 7,851 patients ICA/variance; 973 patients variance control; 492 patients entropy control)

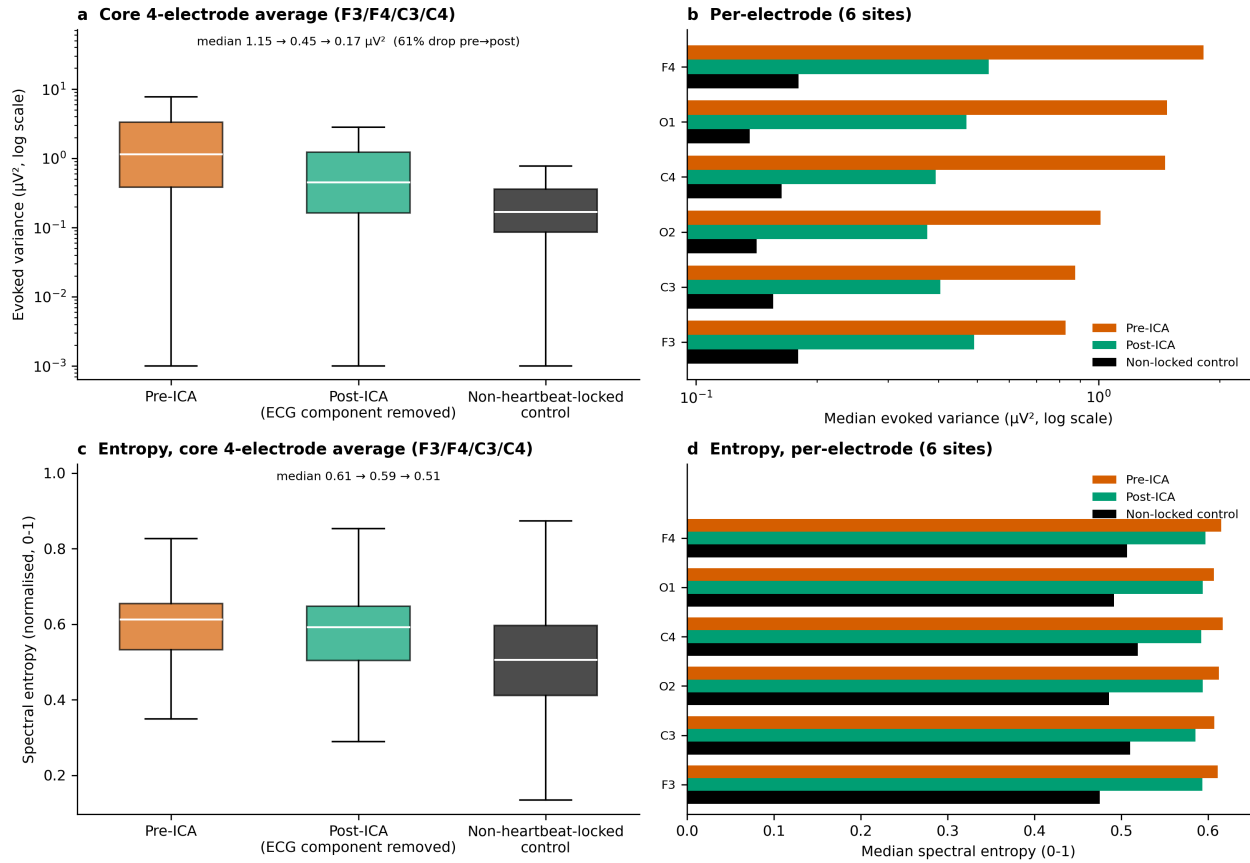


Figure S2. Absolute HEP variance before vs. after excluding the ECG-related ICA component, alongside a non-heartbeat-locked control (n = 7,851 ICA patients with at least one well-covered site; 973-patient control subsample, §2.6; distributions are right-skewed, so medians on a log axis are reported rather than means). (a) Pooled over channels at the four core frontal-central electrodes (F3/F4/C3/C4): median 1.15 μV^2 pre-ICA to 0.45 μV^2 post-ICA (a 61% drop) vs. 0.17 μV^2 for the non-locked control (the same windows re-epoching around random pseudo-events instead of R-peaks). (b) The same three-way comparison broken out per electrode, all six sites (§2.5). (c) Spectral entropy (normalised Shannon entropy of the evoked waveform's Welch power spectrum, 0–1) of the same three conditions, core electrode average (n = 492 patients, separate subsample). (d) Spectral entropy per electrode.

Removal of the ECG-related component reduced median core-electrode HEP variance by 61%; this ratio of group medians is larger than, and not directly comparable to, the median per-channel reduction reported in §3.2 (28%). The non-heartbeat-locked control defines the finite-sample noise floor (0.17 μV^2), which was well below both the pre-ICA (1.15 μV^2) and post-ICA (0.45 μV^2) values. Both conditions therefore retained R-peak-locked structure; post-ICA variance remained above the noise floor at every electrode (panel S2b), indicating that component removal did not reduce the heartbeat-locked signal to noise level. Spectral entropy provided a complementary, variance-independent measure: entropy decreased from pre-ICA (0.61) to post-ICA (0.59) to the non-locked control (0.51); the post-ICA decrease was small, and post-ICA values exceeded the control at every electrode (panel S2d).

S3. EEG-ECG cross-correlation and mutual information

Figure S4. How much of the EEG is heart data? EEG-ECG cross-correlation and mutual information: pre-ICA, post-ICA, and a non-heartbeat-locked control (n = 148 patients)

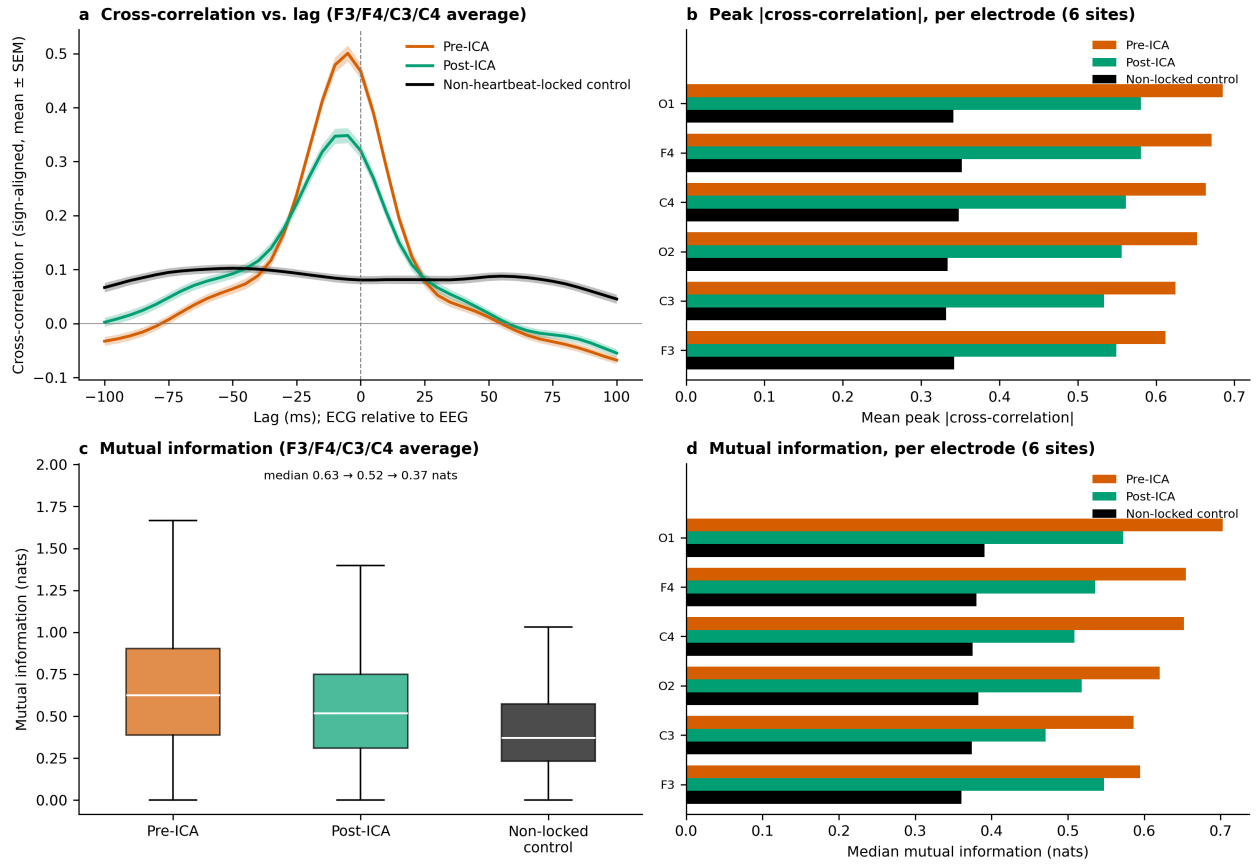


Figure S3. EEG-ECG cross-correlation and mutual information for pre-ICA, post-ICA, and non-heartbeat-locked control conditions (n = 148 patients; separate subsample in which ICA was refitted to obtain paired evoked waveforms; §2.6). In the control condition, the same pre-ICA EEG was re-epoched around random pseudo-events and correlated with the R-peak-locked ECG evoked average, providing the chance-level relationship expected in the absence of R-peak-locked structure. (a) Lag-resolved cross-correlation between the core-electrode (F3/F4/C3/C4) HEP evoked average and the concurrent ECG evoked average, sign-aligned per channel before averaging; mean $\pm$ SEM across patients. (b) Mean peak |cross-correlation| (maximum over lags) per electrode for all three conditions. (c) Mutual information between the same waveform pairs, core-electrode average. (d) Mutual information per electrode.

Cross-correlation peaked near zero lag in both heartbeat-locked conditions (median peak lag -5 ms pre-ICA and -10 ms post-ICA), consistent with the near-instantaneous volume conduction assumed by the zero-lag regression estimator (§2.3). Peak correlation decreased from pre-ICA (mean $|r| = 0.64$) to post-ICA (mean $|r| = 0.56$) to the non-locked control (mean $|r| = 0.34$), and mutual information decreased correspondingly, from 0.63 to 0.52 to 0.37 nats (panel c). Post-ICA values remained above the non-locked floor at every well-covered electrode (panels S3b, S3d), indicating that a non-trivial EEG-ECG relationship persists after cleaning, in agreement with §3.2 and Supplementary Figure S2. Mutual information, which is also sensitive to nonlinear dependence, followed the same ordering as peak correlation, giving no indication of substantial EEG-ECG dependence beyond that captured by the linear measures of §3.2.

S4. Power spectral density: pre-ICA, post-ICA, non-locked control, and ECG

To complement the correlation- and variance-based summaries, we compared the spectral content of each evoked waveform. Welch power spectral density was computed for the pre-ICA, post-ICA, and non-locked evoked waveforms described in §S3 and for the patient's ECG evoked average, in the same 150-patient subsample.

Figure S5. Power spectral density: pre-ICA, post-ICA, non-locked control, and ECG
(n = 150 patients, subsample)

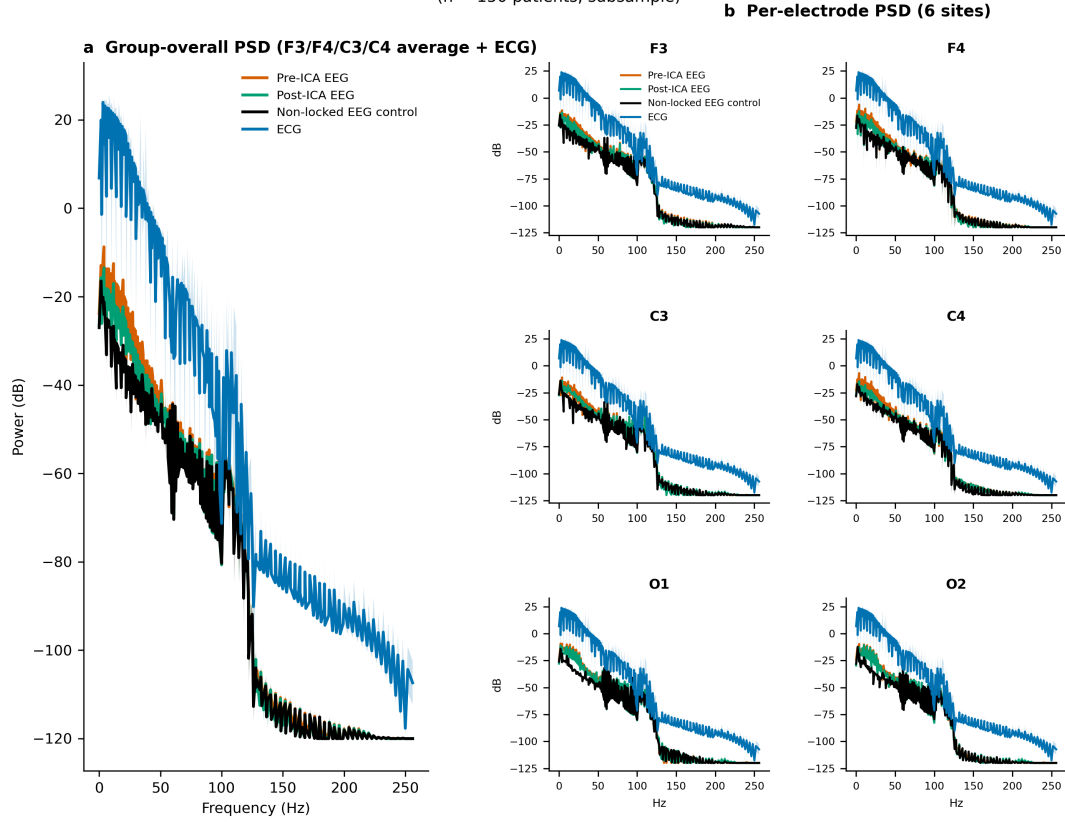


Figure S4. Power spectral density (Welch, dB) of the pre-ICA, post-ICA, non-heartbeat-locked control, and ECG evoked waveforms (n = 150 patients). (a) Group average: core four-electrode (F3/F4/C3/C4) average, all four conditions overlaid (mean $\pm$ SEM). (b) Per-electrode: the same four-way overlay for each of the six well-covered sites (§2.5).

The ECG spectrum was broadband and structured, dominated by the sharp QRS transient, whereas the non-locked EEG control lacked this QRS-dominated structure. Pre-ICA and post-ICA EEG spectra lay between these two references and appeared visually closer to the ECG than the non-locked floor did, providing spectral-domain support for the conclusions of §3.2 and §S3. Spectral shape was visually similar across the six electrodes (panel b), suggesting that CFA varies across the scalp mainly in magnitude (Figure 2) rather than in spectral content.

S5. Adjustment of the sex effect for BMI

Male patients had higher CFA R^2 than female patients (§3.4), whereas mean BMI was higher in female patients (34.8 vs. 31.8 kg/m²); because BMI was itself associated with CFA, the sex comparison could be confounded by BMI. In the subsample with available BMI (n = 6,127), an OLS regression of patient-mean CFA R^2 (outside QRS) on sex alone yielded a male-versus-female coefficient of 0.031 (95% CI 0.020 to 0.041, $p = 1.1\text{e-}08$). Inclusion of BMI as a covariate did not attenuate the sex coefficient (0.040; 95% CI 0.029 to 0.050, $p = 3.2\text{e-}13$), while BMI itself was an independent, significant predictor (coefficient 0.0029 per BMI unit, $p = 1.5\text{e-}21$).

Figure S5. Adjusting for BMI does not shrink the sex coefficient (BMI-available subsample only, underpowered vs. Fig. 4b)

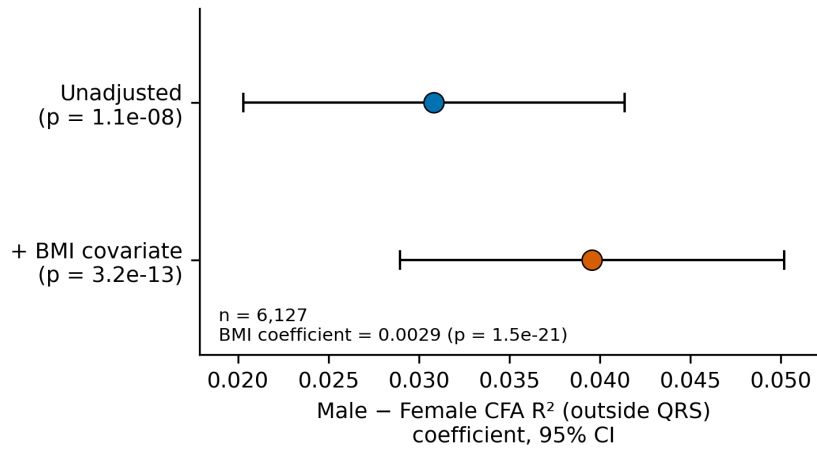


Figure S5. Male-versus-female CFA R² (outside QRS) OLS coefficient, unadjusted vs. BMI-adjusted, same BMI-available subsample (n = 6,127); points = coefficient, error bars = 95% CI.

The sex coefficient did not decrease after adjustment for BMI and, if anything, increased slightly, as expected given the higher BMI of female patients, arguing against BMI as the explanation for the sex effect observed in the full cohort. In this observational model, sex and BMI were thus independently associated with CFA.

S6. Sensitivity to segment duration

Figure S6. Does window length matter? Full-cohort dose-response, n=7,079 common to available lengths

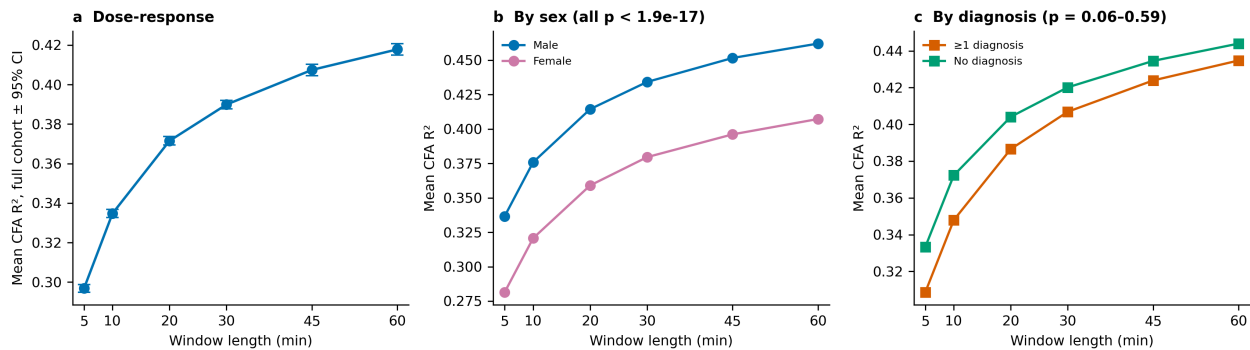


Figure S6. Sensitivity of CFA estimates to segment duration (n = 7,079 patients with usable data at every duration). (a) Mean channel-level CFA R² (all EEG channels) at 5, 10, 20, 30, 45, and 60 min (5 min: 0.30; 10 min: 0.33; 20 min: 0.37; 30 min: 0.39; 45 min: 0.41; 60 min: 0.42). (b) Sex-stratified and (c) diagnosis-stratified estimates. The sex difference was present at every available duration (Mann-Whitney, all p ≤ 1.94e-17); diagnosis differences were not significant (p = 0.06–0.59).

Segment duration had a clear effect on the detectable cardiac contribution. Mean CFA R² increased from 0.30 at 5 min to 0.33 at 10 min, 0.37 at 20 min, 0.39 at 30 min, 0.41 at 45 min, and 0.42 at 60 min, consistent with incomplete averaging of heartbeat-locked structure over fewer beats in shorter windows. The incremental gain decreased from 0.04 R² units between 10 and 20 min to 0.02 between 20 and 30 min and 0.01 between 45 and 60 min, indicating diminishing returns with longer windows, although no plateau was reached within 60 min. Fewer channel-recordings contributed at 45 and 60 min (46,099 vs. 79,304 at 30 min), so the longest-window means may partly reflect a different channel composition; between 5 and 30 min, however, channel composition was nearly constant (79,651 to 79,304 channel-recordings), and R²

still increased monotonically. The sex difference persisted across all tested durations, whereas diagnosis differences were not significant in this matched subset. Unlike in the main analysis, the no-diagnosis mean was numerically higher than the any-diagnosis mean at every duration (e.g. 0.33 vs. 0.31 at 5 min), consistent with the within-site reversal described in the Limitations; the diagnosis association should therefore not be interpreted as a disease effect.

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